PROJECT REPORT

FOR

Foundations of Artificial Intelligence

LEXVIGIL AI LEGAL INTELLIGENCE PLATFORM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report entitled "LexVigil AI Legal Intelligence Platform" is the bonafide work of Rakesh H (240701415) and Sharweshvar P (240701491), who carried out the project work under my supervision.

SIGNATURE OF THE FACULTY INCHARGE

Submitted for the Practical Examination held on _____

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ABSTRACT

Executive summary

LexVigil is an evidence-first contract review platform developed for the Foundations of Artificial Intelligence project. It structures contract text, connects clauses and definitions, tracks amendments, and presents potential inconsistencies alongside their supporting passages. The implemented approach combines document processing, graph-like relationships and bounded symbolic reasoning in a React and FastAPI application.

A synthetic services agreement demonstrates a payment period amended from 30 to 45 days while a late-payment charge remains tied to 30 days. The report links this candidate inconsistency to four source clauses. Thirteen focused engine tests passed during verification on 26 September 2026. These checks establish specific software behaviours, not legal accuracy across unseen contracts. Local analysis is available; live AI review and research requests remain disabled. Broader evaluation and deployment hardening are required before production use.

1 INTRODUCTION

Contract obligations may depend on definitions, cross-references and later amendments. Reviewing isolated sentences can miss these relationships. LexVigil provides a workspace where users can inspect a potential issue and trace the underlying text before reaching a conclusion.

The project objectives are to:

- Extract numbered clauses and preserve source locations.
- Resolve explicit references and identify bounded inconsistency patterns.
- Support human review through evidence navigation, saving and export.

2 LITERATURE REVIEW

Hendrycks et al. (2021) introduced CUAD, an expert-annotated contract dataset covering 41 clause categories. Koreeda and Manning (2021) introduced ContractNLI, pairing document-level inference with supporting evidence spans across 607 contracts. These studies motivate structured extraction and evidence presentation. LexVigil uses deterministic rules rather than a model trained on either dataset; neither study establishes its performance. A future benchmark should assess clause extraction, reference resolution and finding quality separately.

3 PROPOSED SYSTEM

The local development design separates presentation, authenticated processing and persistence. Uploaded documents pass through bounded ingestion before clause parsing and reference analysis. The engine derives candidate findings from explicit patterns and returns source-linked reports. SQLAlchemy manages report persistence, while consent and retention rules govern temporary originals.

The system distinguishes source text from inferred findings. Unresolved references remain visible, and ambiguous interpretations require human review. A reduced hosted showcase has narrower capabilities than the development profile, including deferred OCR and original storage.

3.1 Architecture diagram

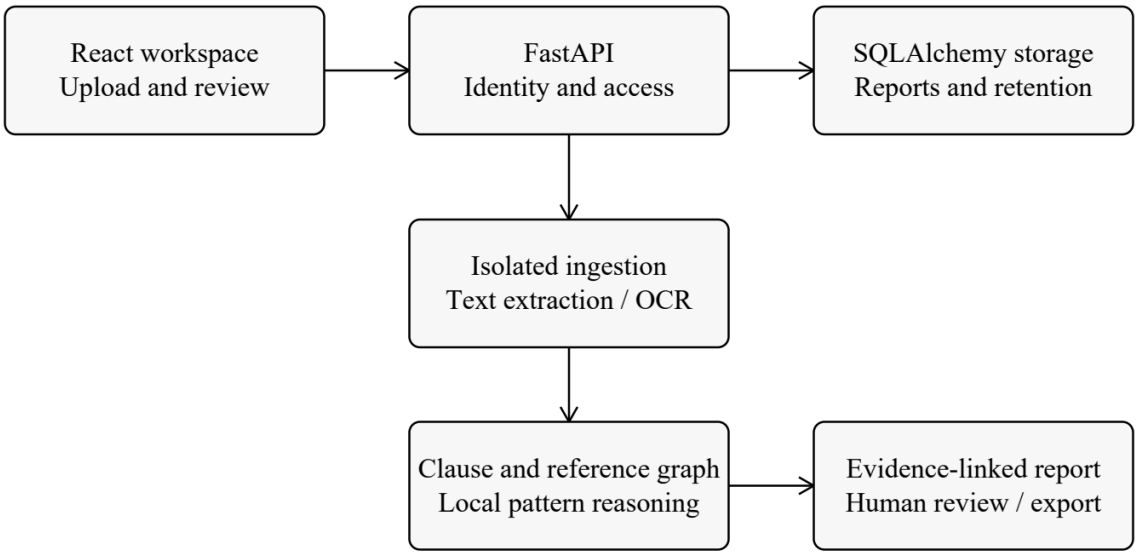


Figure 3.1 Local development architecture and analysis flow

4 MODULES DESCRIPTION

4.1 Ingestion and reasoning

The ingestion module accepts supported PDF, DOCX, TXT and Markdown documents. Development OCR is optional for English, Tamil and Hindi. Parsing identifies clauses, definitions, references and amendments. Pattern rules examine relationships such as effective payment periods and late-payment triggers, retaining the clauses supporting each candidate.

4.2 Evidence and report management

The workspace displays findings, source clauses and coverage limitations. Users can save, export or delete analyses. Unsaved reports normally expire after 24 hours. Consented development originals have a separate maximum 48-hour lifetime; saving a report does not extend it.

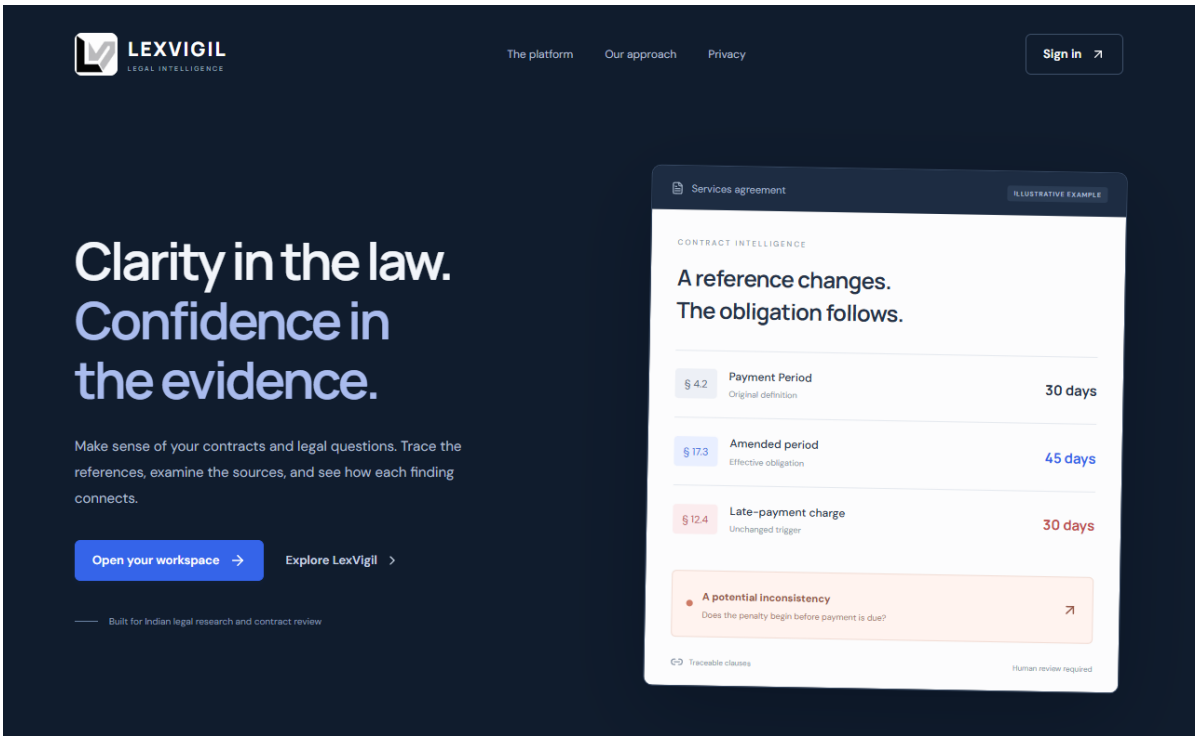


Figure 4.1 Landing page hero from the saved 13 September 2026 demonstration

5 IMPLEMENTATION AND RESULTS

5.1 Experimental setup

The implementation uses React 19, TypeScript, FastAPI, SQLAlchemy and local SQLite persistence. Verification ran the existing contract-engine test module on Windows. The synthetic agreement supplies controlled clauses; it is not a representative legal benchmark.

5.2 Results and analysis

All 13 focused tests passed. The sample contains six clauses and one temporal finding. Its 45-day amended payment period conflicts potentially with a 30-day penalty trigger, subject to matching invoice scope. The evidence chain connects clauses 4.2, 8.1, 17.3 and 12.4.

- Future amendments are assessed against the selected analysis date.
- Missing references remain unresolved rather than receiving guessed targets.
- Explicit termination exceptions are distinguished from simple conflicts.
- Invalid inputs are rejected; PDF and DOCX ingestion are checked.

These results support traceable, bounded behaviour. They do not measure precision, recall, review-time savings or legal correctness. The saved screenshot on the next page illustrates the sample output; the current test run independently verifies its central engine behaviour.

5.3 Application screenshot

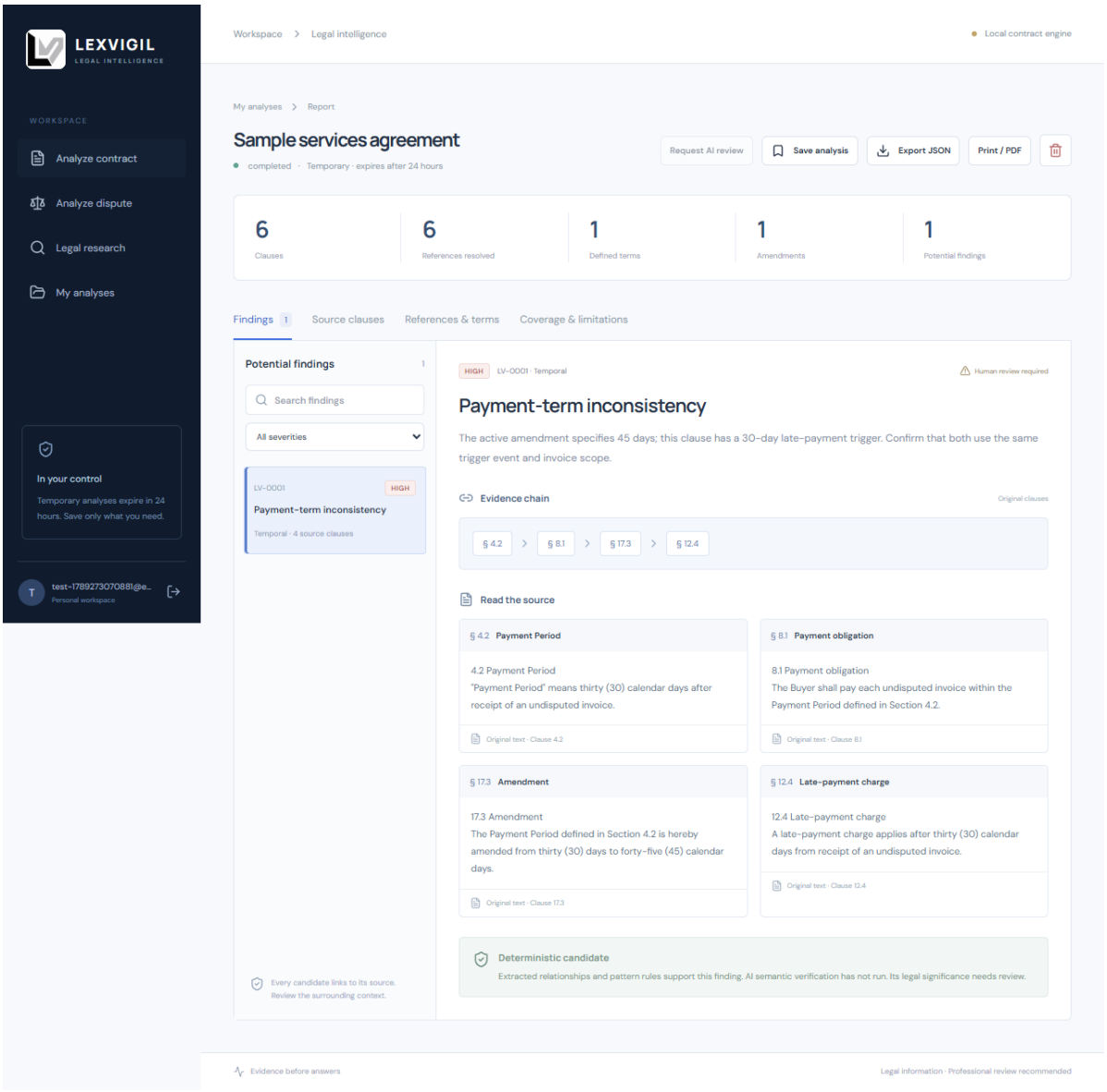


Figure 5.1 Sample contract finding and evidence chain, saved 13 September 2026

The report places the potential inconsistency beside its source clauses and labels human review as required. The sample is synthetic.

6 CONCLUSION AND FUTURE WORK

LexVigil demonstrates how structured extraction and symbolic reasoning can make contract findings inspectable. Its contribution is a traceable review workflow with explicit limits. The next priorities are expert-labelled evaluation, broader layout and language testing, and deployment validation. Live AI functionality should be enabled only after an eligible India-only processing route is integrated and verified. Professional review remains necessary.

7 REFERENCES

Hendrycks, D., Burns, C., Chen, A. and Ball, S. (2021). CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. arXiv:2103.06268. https://arxiv.org/abs/2103.06268

Koreeda, Y. and Manning, C. (2021). ContractNLI: A Dataset for Document-level Natural Language Inference for Contracts. Findings of EMNLP, pp. 1907-1919. https://aclanthology.org/2021.findings-emnlp.164/

LexVigil project repository (2026). README; contract engine; release scope; engine tests; saved demonstration screenshots. Reviewed 26 September 2026.